

Public CAD Is Not Operational Ground Truth

Hurricane Ida and the Measurement of Police Responsiveness

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Abstract

Public computer-aided dispatch (CAD) records are often used to measure emergency-service responsiveness, but their timestamps and dispositions are outputs of both operational activity and a recording, retention, and export process. This paper documents that distinction using Hurricane Ida as a system-level stress test of New Orleans public CAD. The all-call Ida score interval lies above the score intervals of all 217 qualified reference windows and remains separated in two narrower prespecified reference designs. The released dispatch–arrival field configuration changes across call-creation cohorts and within recorded dispositions; among arrival-observed records, the late shift is not explained solely by the observed final-disposition mix. A frozen timing-shift placebo is unfavorable, so the evidence does not identify precise institutional timing or a mechanism. These findings establish a replicated, reference-extreme change in the released measurement product, not a change in physical police response or domestic-violence incidence. The paper then uses standard partial-identification logic to show what the public record can support directly and which additional clocks, histories, coverage measures, and retained lineage are needed for performance measurement. Reported domestic-violence-related calls provide a policy-relevant application. The estimates remain system-level; the application defines the evidence required for a future DV performance study rather than reporting one.

Keywords: calls for service; police performance measurement; domestic violence; Hurricane Ida; measurement validity; partial identification

JEL codes: C18, C55, H12, K42

1 Introduction

Public managers and researchers routinely use computer-aided dispatch records to describe emergency-service demand and responsiveness. The appeal is clear: the files are large, timestamped, and often publicly reproducible. The difficulty is that a released timestamp is not an operational event by definition. It is an administrative field whose meaning depends on how activity was recorded, retained, and exported.

Workload theory organizes the operational side of this problem. Let D_t^{CAD} denote reported demand entering CAD, C_t effective capacity, and P_t the priority process. The resulting service path is

$$S_t = g(D_t^{\text{CAD}}, C_t, P_t). \quad (1)$$

Neither true need nor attempted help-seeking is identical to this CAD denominator. The demand chain runs from latent need D_t^* through help-seeking attempts H_t and contacts received by the emergency system A_t to calls created in CAD, D_t^{CAD} . The paper studies performance conditional on that last population.

Public CAD is a tempting way to observe S_t : it is large, timely, and granular. Yet every released timestamp, priority, and disposition also passes through a recording, retention, and export process R_t . Researchers observe

$$Y_t = h(S_t, R_t), \quad (2)$$

and a candidate performance measure is $\theta_k = m_k(S)$. The relevant identified set contains the values of θ_k generated by operational and record-production paths compatible with the released data and admitted witnesses. Its width determines whether a measure is directly supportable, bounded, selected, or unavailable.

Hurricane Ida establishes why this observation layer matters empirically. Access to 911, police operations, and record production were strained together. The analysis asks whether relationships among released fields remained within a frozen reference class. It finds a reference-extreme, time-structured, within-disposition change in the public trace. The stress test cannot assign that change to service or recording, which is precisely the ambiguity that a performance measure must resolve.

Reported domestic-violence-related calls supply a demanding downstream application. For those calls, managers and civilian reviewers may want to know how long calls waited, how priorities were revised, which qualified endpoint was reached, and whether required follow-up occurred. Each quantity is meaningful only after its call denominator, label, clock, endpoint, and lineage are established.

The paper’s central argument is:

Hurricane Ida reveals a replicated break in the stability of the released CAD measurement product. Performance interpretation therefore requires measure-specific evidence beyond public field presence.

Three connected contributions implement this argument. First, the empirical module constructs four dispatch–arrival field-presence states, standardizes them across ten frozen twelve-hour bins, displays the complete 217-window reference distribution, and localizes the Ida movement within recorded dispositions. Second, the measurement module separates operational activity from record

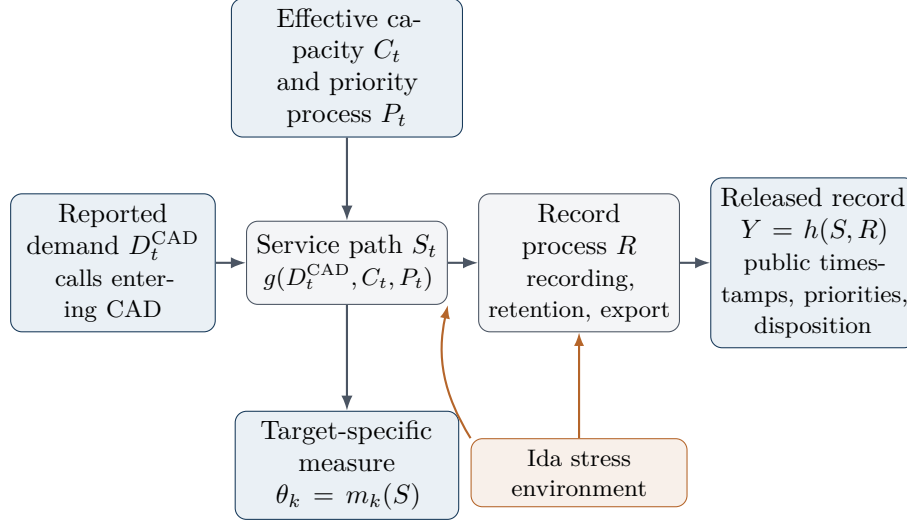


Figure 1. Workload produces the service path; the record process produces the public observation. Target-specific witnesses establish which service measures the observation supports.

production and maps each performance target into the set of service paths consistent with the public record. Third, a concise reported-DV application follows classification, response endpoints, priority histories, EPR production, and follow-up to specify the aggregate evidence a later empirical study would require.

Each element has a distinct role. Existing research develops police-performance targets, workload relationships, administrative-data error, and partial-identification methods. The contribution here is empirical and design-oriented rather than a new estimator: a reproducible diagnostic for the shared CAD observation layer, followed by target-specific identified sets and privacy-conscious aggregate release specifications. This extends the police-workload perspective associated with Beland (2026). Reported demand and effective supply shape service, while the administrative observation layer determines which workload implications can be measured.

The result is constructive for both management and oversight. Better lineage, clocks, and histories contract identified sets directly. They can support future allocation models and sharper accountability measures on a common reported-call denominator. Estimating a DV performance level, the effect of a reform, or a behavioral mechanism remains a subsequent empirical task with its own classification authority and research design.

2 Economic and empirical background

Police performance is multidimensional, and administrative convenience is a weak basis for choosing a metric (Moore and Braga, 2004; Tiwana et al., 2015). More generally, public-sector metrics can redirect effort toward recorded targets and alter the records used for accountability (Courty and Marschke, 1997; Eckhouse, 2022). For domestic and family violence, Rollings and Taylor (2008) propose measures of recording accuracy, repeat attendance, repeat offending, charging, information supplied to responders, and victim satisfaction. EIGE likewise organizes police and justice data around thirteen IPV indicators and comparable administrative definitions (European Institute for Gender Equality, 2023). These studies define substantive targets. The present paper asks which targets can be recovered from a public CAD release.

Responsiveness illustrates the issue. Faster police arrival can affect clearance and injury outcomes (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023). Using more than 2.7 million California fire-service incidents, Brent and Beland (2020) show that traffic congestion slows first responders and increases damages; their linked operational data make clear how economically consequential a valid response clock can be. Yet response time varies across reporting, dispatch, travel, and combined clocks, and the literature uses heterogeneous priority definitions and call populations (Verlaan and Ruiter, 2023). Rich internal data can link staffing, call volume, proactive work, priority, and response times (Mourtgos et al., 2024). A public release that omits the queue, availability, or a qualified endpoint supports a narrower statistic.

The DV literature further separates reported demand from the underlying phenomenon. Calls, incidents, arrests, hotline contacts, and hospital records can move differently during the same disruption (Leslie and Wilson, 2020; Bullinger et al., 2021; Miller et al., 2022, 2024). Disaster studies using community, postpartum, displaced-population, and adult survey samples report post-disaster associations among hurricane exposure, partner conflict or violence, and mental-health outcomes (Anastario et al., 2009; Schumacher et al., 2010; Harville et al., 2011; First et al., 2022). Those studies motivate the substantive importance of a disaster-period DV measurement design, but they do not validate public CAD timestamps, identify an *Ida* effect on reported DV calls, or resolve selection into police records. Police composition can affect reporting (Miller and Segal, 2019), while stable identifiers permit linkage from police reports to later economic outcomes (Adams et al., 2024). This paper therefore targets performance conditional on a reported DV-related call entering CAD.

Administrative records reflect selection, classification, and linkage. Reporting, coding, and processing create successive gates (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021; Knox et al., 2020). Call-takers assign categories and priorities before resources are deployed (Pandey, 2025). Fragmented DV systems and inconsistent identifiers complicate links among incidents, reports, referrals, and follow-up (Phoenix, 2023; Vomfell and Povala, 2023). Changes in incentives can alter registered crime (Ahmed et al., 2026). The measurement design must therefore describe how an operational stage becomes an observed record.

Econometric work provides the corresponding identification logic. Classical identification asks whether the observable distribution distinguishes the underlying structure (Koopmans and Reiersøl, 1950; Rothenberg, 1971; Lewbel, 2019). Partial identification retains the complete set of observationally equivalent target values when it does not (Manski, 1989, 1990, 2003; Molinari, 2020). Measurement error in nonlinear settings generally requires auxiliary measurements, instruments, validation data, or explicit restrictions (Chen et al., 2011). Administrative sources can measure a different concept or contain their own linkage and classification errors (Kapteyn and Ypma, 2007). Dominitz and Sherman (2006) derive sharp bounds on school-performance measures when an administrative verification indicator is imperfect; their measure-specific treatment of validity is especially close to the present application. Communicating the resulting nonsampling and conceptual uncertainty is part of valid official measurement (Manski, 2015).

Two further connections organize the witness analysis. Blackwell (1953) compares information structures by the decisions they support, while Masten and Poirier (2020) locates boundaries at which conclusions cease to survive relaxed assumptions. Here, each administrative witness refines the feasible set of operational paths for a declared performance target. Sufficiency is reached when the target’s identified set meets a prespecified precision; one-at-a-time removal establishes whether

every retained witness is necessary. The resulting inclusion-minimal bundle is a target-specific release design rather than a universal ranking of data fields.

Economists also use disasters to reveal system dependencies that routine variation leaves hidden. The Great East Japan Earthquake exposed propagation through directly observed firm networks (Carvalho et al., 2021); hurricanes and outages reveal payment resilience by combining transactions, outages, surveys, and experimental evidence (Alvarez et al., 2026). Environment comparisons can also test whether a statistical relationship remains invariant, although causal interpretation requires additional structural assumptions (Peters et al., 2016). BELAND-PLUS uses Ida at the diagnostic stage: the event tests stability of the released record, while operational witnesses determine the performance measure.

Partial identification supplies the formal response. Missing outcomes yield sharp regions, restrictions on misclassification can tighten them, and additional evidence or restrictions contract the set of observationally equivalent values (Manski, 1990, 2003; Horowitz and Manski, 1995; Molinari, 2008, 2020; Tamer, 2010). BELAND-PLUS applies that logic to the measurement stage itself. Ida establishes the empirical relevance of the problem; the DV chain defines the candidate functionals and the witnesses that contract their identified sets.

3 Institutional setting and research design

3.1 Hurricane Ida as a system-level stress test

Hurricane Ida made landfall in Louisiana on 29 August 2021. The Office of the Independent Police Monitor documented twelve-hour police shifts, a period when 911 dispatch was down, anti-looting deployments, and a citywide curfew during the event (Office of the Independent Police Monitor, 2021). Reporting, police operations, and record production were therefore exposed to simultaneous stress. This is precisely what makes Ida informative as a measurement test: it can reveal whether relationships among public fields remain stable when the system producing them is under unusual strain.

Ida is the test bed. The design uses its documented operational conditions to probe the public measurement layer; the observed topology remains compatible with several operational and record-production pathways.

The chronology is post-anchor prospective, not a pre-Ida preregistration. Ida was selected as the focal stress-test anchor before this reference-calibration stage; the three reference universes, estimator, eligibility rules, thinning rule, interval optimization, placebos, and stop rules were then frozen before the pseudo-event reference outcomes were examined.

3.2 Why an all-call stress test informs DV measurement

The identified-set logic can be written from the schema alone. Ida adds the empirical premise that makes the exercise consequential: the shared public observation layer actually moved outside its frozen reference class under system stress. The result replaces a generic warning about administrative data with a measured failure of stability in the fields later used for crime-specific performance analysis.

The empirical analysis uses all reported calls because every downstream call-type measure inherits that CAD layer. A DV labeling rule changes the analytic population, while its priorities, timestamps, dispositions, and links still pass through the tested record-production system. The transfer is

Table 1. Public field-presence states

State	Dispatch field	Arrival field	Safe interpretation
J_{00}	absent	absent	neither public field is present
J_{01}	absent	present	arrival field is present; dispatch field is absent
J_{10}	present	absent	dispatch field is present; arrival field is absent
J_{11}	present	present	both public fields are present

These states record public field presence. Physical-stage status requires a separate operational bridge.

therefore a measurement requirement. Each DV measure must establish its denominator, labeling rule, clock, endpoint selection, and lineage. The system-level estimates supply no DV-specific magnitude.

3.3 Making the public record’s shape measurable

The analysis uses the official 2021 New Orleans Calls for Service file (City of New Orleans, 2021). The denominator is the frozen public-created-call universe used throughout the project. For record i , define

$$D_i = \mathbf{1}\{\text{a valid public dispatch field is present}\}, \quad (3)$$

$$A_i = \mathbf{1}\{\text{a valid public arrival field is present}\}. \quad (4)$$

The two indicators generate four mutually exclusive public record states:

Exactly one state is realized for every record, so the state shares sum to one. This simplex identity is central: when the share of J_{11} falls, the mass must move to one or more other public states. The analysis is therefore about redistribution within the public record, not simply about a count declining.

3.4 Event window and time bins

The Ida window runs from 29 August 2021 at 00:00 through 3 September at 00:00 local time. It is divided prospectively into ten twelve-hour bins. To avoid unexplained internal labels, the paper reports calendar time directly; the technical labels are retained only for reproducibility. Each bin indexes when a cohort’s call record was created; field states are taken from the eventual released record. The figures are therefore not contemporaneous snapshots of physical dispatch or arrival at that clock time.

Table 2. Calendar interpretation of the focal bins

Technical bins	Calendar interval	Role in the descriptive result
B1	29 Aug., 00:00–12:00	opening interval; not among the nine threshold-exceeding cells
B2–B5	29 Aug., 12:00–31 Aug., 12:00	earlier pattern: lower J_{11} share
B6–B7	31 Aug., 12:00–1 Sep., 12:00	later pattern: lower J_{11} and higher J_{01}
B8	1 Sep., 12:00–2 Sep., 00:00	negative J_{11} tail
B9–B10	2 Sep., 00:00–3 Sep., 00:00	no cell exceeds the common threshold

3.5 Standardized event–reference contrasts

Within stratum s , time bin t , and state j , let p_{jst}^E and p_{jst}^R denote the event and reference shares. Using the frozen reference-derived common-support weights ω_{st}^R , the standardized event–reference contrast is

$$\Delta_{j,t} = \sum_{s \in \mathcal{S}_t} \omega_{st}^R (p_{jst}^E - p_{jst}^R), \quad \sum_{s \in \mathcal{S}_t} \omega_{st}^R = 1. \quad (5)$$

Because the four state shares sum to one,

$$\Delta_{00,t} + \Delta_{01,t} + \Delta_{10,t} + \Delta_{11,t} = 0. \quad (6)$$

The primary topology matrix keeps the three independent coordinates $(\Delta_{01,t}, \Delta_{10,t}, \Delta_{11,t})$ for the ten bins.

3.6 Benchmarking rarity against frozen reference windows

The topology score measures how closely any authorized restricted-support representation can approximate the full three-coordinate path while respecting its support envelope. Let G be the 10×3 contrast matrix and $\mathcal{X}$ the frozen finite set of authorized support triples. The full score can be written compactly as

$$U_{\text{full}}(G) = \min_{X \in \mathcal{X}} \max_{j \in \{01,10,11\}} \min_{\beta_j} \|g_j - X\beta_j\|_{\infty}, \quad (7)$$

subject to the rowwise capacity-envelope restrictions defined in the mathematical appendix. A larger score means that the best authorized representation leaves a larger maximum cell discrepancy. The score is a geometric measure of public-record fit.

The same estimator is applied to every qualified reference window. We report interval separation, the complete reference distribution, and the add-one reference fraction $r/(R+1)$ as descriptive measures of reference rarity.

4 Empirical results

The stress test must establish more than an unusual average. It asks whether Ida lies outside the range of the same measurement applied to qualified reference windows, whether the public-record shape changes during the event, and whether the later movement is merely a change in recorded-disposition mix. The three results below answer those questions in sequence.

4.1 Ida is reference-extreme

Ida’s locked score interval is

$$U_{\text{full}}(G_{\text{Ida}}) \in [0.250734484671, 0.250734494671]. \quad (8)$$

No reference interval overlaps it. Ida’s score exceeds the score intervals of all 217 full qualified reference windows, all 153 stage-era-matched windows, and all 86 same-season-stage windows. The corresponding add-one reference fractions are $1/218$, $1/154$, and $1/87$.

The largest reference upper endpoint is approximately 0.2125, leaving a minimum interval separation of about 0.0382 from Ida. This separation is more informative than the narrower fact that Ida lies just above the frozen M7A restricted-support exclusion threshold of 0.25. The threshold

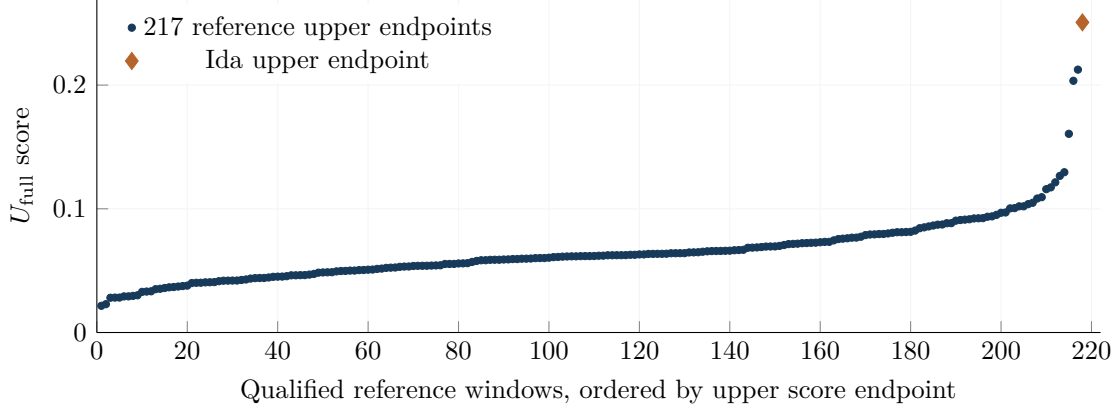


Figure 2. Complete distribution of the 217 full-qualified reference score upper endpoints and the Ida upper endpoint. The narrow score intervals do not overlap: the largest reference upper endpoint is 0.2125162, below the Ida lower endpoint 0.2507345.

is a prespecified model-fit boundary; the empirical rank comparison is invariant to it.

4.2 The timing placebo does not support precise institutional alignment

The placebos distinguish topological rarity from a timing or mechanism claim. Against three frozen unlabeled duration- and mass-matched patterns, the aligned structural library has an advantage interval of $[0.2045072115, 0.2045072215]$. In the separate zero-padded shift exercise, however, the Ida alignment-advantage interval is $[-0.1137314286, -0.0975331874]$: at least one arbitrary translation fits better than the documented alignment. The evidence therefore supports an unusual released-record shape, but not precise alignment with the curfew, outages, or another institutional mechanism.

4.3 The eventual public-state configuration changes across later call-creation cohorts

A cell is classified as unusually large when its absolute contrast exceeds the 95th percentile of the reference-window maximum absolute cell contrast,

$$c_{0.95} = 0.209397476359. \quad (9)$$

Nine bin-state cells exceed this common threshold. The pattern is easier to understand in calendar time than through technical bin labels:

- From the afternoon of 29 August through the morning of 31 August, the abnormal cells are reductions in J_{11} , the state in which both public fields are present.
- From the afternoon of 31 August through the morning of 1 September, J_{11} remains sharply lower, while J_{01} —arrival present but dispatch absent—becomes sharply higher.
- During the afternoon and evening of 1 September, the negative J_{11} contrast remains above the threshold, but J_{01} no longer does.

The result is a redistribution statement: one public state falls while another rises. Its generating pathway remains set-valued.

J_{01} : arrival present, dispatch absent	–	–	–	–	+0.307	+0.323	–
J_{11} : both fields present	-0.422	-0.354	-0.216	-0.266	-0.507	-0.455	-0.228
	Aug 29 PM	Aug 30 AM	Aug 30 PM	Aug 31 AM	Aug 31 PM	Sep 1 AM	Sep 1 PM

Figure 3. The nine threshold-exceeding public-record cells. Values are event-minus-reference probability-point contrasts; a dash marks a cell below the common threshold. The figure describes public field configurations.

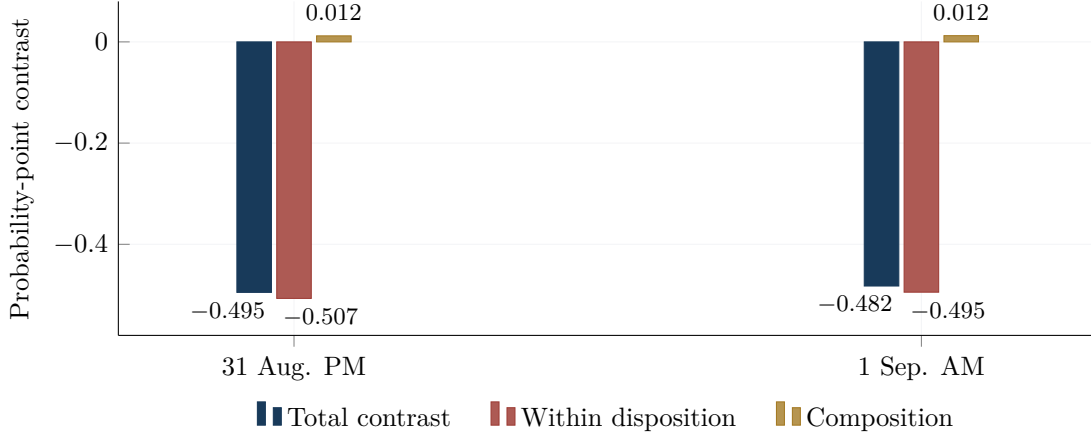


Figure 4. Exact accounting decomposition of the late aggregate shift. The small positive composition term has the opposite sign from the negative total contrast.

4.4 The later shift occurs within recorded dispositions

The next question is whether the late pattern could arise solely because the mix of dispositions changed. Restrict attention to arrival-observed records, $J_{01} + J_{11}$. For disposition r and population $z \in \{E, R\}$, let

$$q_r^z = \Pr(D = 1 \mid A = 1, R = r, z), \quad \pi_r^z = \Pr(R = r \mid A = 1, z). \quad (10)$$

The aggregate dispatch-field share among arrival-observed records is $q_{\text{all}}^z = \sum_r \pi_r^z q_r^z$. The exact symmetric Kitagawa identity is

$$\begin{aligned} q_{\text{all}}^E - q_{\text{all}}^R &= \sum_r \frac{q_r^E + q_r^R}{2} (\pi_r^E - \pi_r^R) \\ &\quad + \sum_r \frac{\pi_r^E + \pi_r^R}{2} (q_r^E - q_r^R). \end{aligned} \quad (11)$$

The first term is the recorded-disposition composition component. The second is the within-disposition component.

For 31 August afternoon/evening, the total contrast is -0.4950 , the composition component is $+0.0119$, and the within-disposition component is -0.5069 . For 1 September morning, the corresponding values are -0.4824 , $+0.0123$, and -0.4946 . The within-disposition component

accounts for the direction and almost all of the magnitude of the aggregate shift in this decomposition.

The sign pattern strengthens the accounting result. For Gone on Arrival (GOA), Report to Follow (RTF), and Necessary Action Taken (NAT), $\Delta H_{01,r} > 0$ while $\Delta H_{11,r} < 0$ in both late bins. A composition-only explanation based on the observed final-disposition mix among arrival-observed records requires matching signs, so the six prespecified cells rule out that explanation alone. This does not rule out selection into the arrival-observed set, unobserved composition, within-category heterogeneity, semantic drift, or alternative service and record-production pathways.

Table 3. Within-disposition change in the public dispatch-field share among arrival-observed records

Calendar bin	GOA	RTF	NAT
31 Aug., 12:00–24:00	−0.440	−0.567	−0.532
1 Sep., 00:00–12:00	−0.265	−0.459	−0.649

Each value is an event-minus-reference contrast in the public dispatch-field share among arrival-observed records within the named recorded disposition. GOA is Gone on Arrival; RTF is Report to Follow; NAT is Necessary Action Taken. All three are administrative categories.

Table 3 reports conditional-share contrasts, whereas Appendix Section 5.1 factors the corresponding joint masses as $H_{11} = mq$ and $H_{01} = m(1 - q)$; Appendix Table 4 reports their contrasts. The tables therefore describe linked but distinct quantities: the first localizes conditional public-field movement, while the second supplies the sign witnesses used to exclude composition-only.

Together, the results establish an extreme, time-structured change in the public administrative trace, including movement within recorded dispositions. They show that relationships among released fields were not mechanically stable during Ida; they do not establish which operational or record-production pathway changed.

This is the empirical bridge to DV measurement. A DV label selects records after they have passed through the same field-production system. The performance question then becomes measure-specific: which operational functionals retain a unique interpretation after that selection?

The all-call result neither estimates DV demand nor implies that every DV measure fails. It establishes a common observation-layer problem. The DV application determines which targets survive that problem and which require additional evidence.

Stress-test implication. Ida establishes instability in the released measurement product. Operational interpretation requires a separate identified-set analysis for each downstream measure.

5 From the Ida diagnostic to measure-specific identification

The empirical statistic and the performance target are different mathematical objects. The stress-test statistic is a function of released records,

$$T(Y) := U_{\text{full}}(G(Y)). \quad (12)$$

Its reference rank establishes that Ida’s public-field path is unusual within the frozen comparison design. A candidate performance target instead depends on the latent operational path S .

Let W denote the available semantic and operational witnesses, and let $\mathcal{C}(W)$ collect the paths

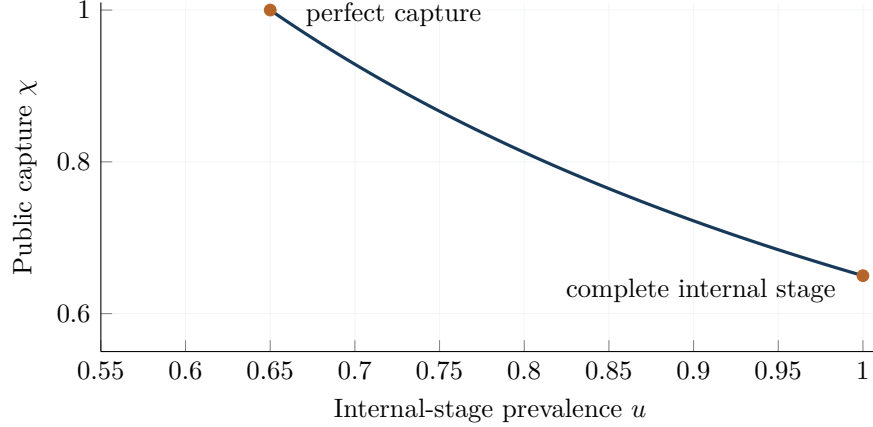


Figure 5. Illustrative product identified set for $q^{\text{pub}} = 0.65$. Observing the product selects a curve of internal-stage and capture values. The value 0.65 is illustrative.

and recording processes consistent with those witnesses. For candidate measure $\theta_k = m_k(S)$, define

$$\mathcal{I}_k(Y, W) = \{m_k(S) : Y = h(S, R), (S, R) \in \mathcal{C}(W)\}. \quad (13)$$

The measure is directly supportable when this set is a singleton, bounded when it is informative but non-singleton, selected when its observed version conditions on a narrower risk set, and unavailable when current evidence leaves the target unrestricted or undefined. New witnesses contract $\mathcal{C}(W)$ and therefore weakly contract $\mathcal{I}_k(Y, W)$.

5.1 Internal stage occurrence versus public capture

Suppose future semantic evidence established a valid direct bridge between a qualified internal administrative dispatch event and the public dispatch field. Let

$$q^{\text{pub}} = \Pr(D_{\text{pub}} = 1 \mid A_{\text{pub}} = 1, R = r, t), \quad (14)$$

$$u = \Pr(U_{\text{dispatch}} = 1 \mid A_{\text{pub}} = 1, R = r, t), \quad (15)$$

$$\chi = \Pr(D_{\text{pub}} = 1 \mid U_{\text{dispatch}} = 1, A_{\text{pub}} = 1, R = r, t). \quad (16)$$

Under exact nesting, the observed public share factorizes as

$$q^{\text{pub}} = u\chi. \quad (17)$$

For any $0 < q^{\text{pub}} < 1$, the identified set is

$$\mathcal{I}(u, \chi \mid q^{\text{pub}}) = \{(u, q^{\text{pub}}/u) : u \in [q^{\text{pub}}, 1]\}. \quad (18)$$

The same public value is compatible with perfect public capture and a lower internal-stage prevalence, complete internal-stage prevalence and imperfect public capture, and every intermediate combination.

The product identifies a curve rather than a unique pair. Joint public-internal counts would separate internal-stage prevalence from public capture. This example is a special case of equation (13): timestamp precision does not establish the mapping from a public field to the operational stage it is

intended to measure.

A DV labeling rule restricts the analytic population but leaves the observation map unchanged. The DV results below are derived by applying equation (13) to that selected chain. They are measurement results, not an empirical DV subset analysis.

5.2 Queue, availability, priority history, and continuity

Queue dynamics live on a different denominator. A qualified stock-flow identity would have the form

$$Q_{t+1} = Q_t + I_t - O_t - X_t, \quad (19)$$

where Q_t is calls holding, I_t is inflow, O_t is ordinary outflow, and X_t is another qualified exit. Populating the identity requires calls-holding stocks, inflows, exits, and realized availability. Ordered priority histories and follow-up states add the remaining transitions.

Endpoint timestamps omit the stock waiting, the resources available, and the sequence of priority and exit changes. The product and queue examples show how each performance target selects its own denominator, clock, and witness requirements.

5.3 Operational and record-production friction

The observation map separates two forms of friction. Operational friction changes the service path through congestion, unit availability, assignment, handoffs, or follow-up. Record-production friction changes how those stages are classified, overwritten, retained, linked, or exported. Both can produce gaps in public timestamps or changes in public-state composition.

The Ida pattern is compatible with either source, or with both operating together. The diagnostic supplies no probability ordering over them. Target-specific witnesses localize the ambiguity: holding and availability histories recover operational congestion; joint public-internal counts recover capture; retained lineage reveals where records cease to share a denominator. A better release therefore improves measurement by contracting the relevant identified set. Whether a reform also reduces operational friction is a separate evaluative question.

6 From identified sets to admissible measures

Equation 13 turns five design choices into an admissibility test: labeling rule, denominator, operational clock, endpoint selection, and stage lineage. Equation 17 motivates lineage; Equation 19 motivates stocks and transitions. Ida makes these choices empirically consequential because the shared public trace changed while its generating pathway remained set-valued.

Directly supportable administrative measures are functions of the released file under an explicit rule and denominator. Bounds propagate unresolved labels or missing outcomes. Selected measures describe the risk set on which their endpoints are observed. Measures requiring absent clocks, states, or links become available when the corresponding witnesses enter W .

Responsiveness requires a time origin, endpoint, and population at risk. Prioritization requires an ordered history when priorities change. Follow-through requires links from CAD to the applicable downstream stage, such as EPR production, program referral, or a policy-defined callback. A recorded disposition can mark an administrative branch, including GOA, while later duties require their own evidence.

NOPD policy anticipates assigned units, dispatch, arrival, return-to-service, and disposition in

the communications record (New Orleans Police Department, 2021). Later oversight and court-directed reforms identify response time, calls holding, priority changes, officer availability, callbacks, accessibility, and data quality as operational objects (City of New Orleans and U.S. Department of Justice, 2024; U.S. District Court for the Eastern District of Louisiana, 2025; Office of the Independent Police Monitor, 2022, 2026). These documents establish why each object matters, while the public CAD files do not record callback eligibility or completion. Empirical measurement requires period-specific values and retained lineage.

Before applying the framework to reported DV-related calls, one further question is necessary: did later public-data enrichment supply the missing clocks and links, making the Ida lesson specific to the 2021 data regime?

7 Do newer public data close the measurement gap?

We retrieved and froze the official 2025 and 2026 Calls for Service datasets, their metadata and dictionaries, the current response-time dashboard model, current response policies, and the 2025 Electronic Police Report metadata (City of New Orleans, 2025, 2026; New Orleans Police Department, 2025, 2024, 2026). The comparison evaluates public measurement regimes.

The newer regime adds machine-queryable derived fields, including a candidate en-route duration and multiple dispositions. Current policies also clarify priority ordering, alternate response, policy-defined callbacks, and return-to-queue rules. The annual raw schema retains the 2021 structure: initial and recorded signal/priority endpoints, create/dispatch/arrival/closure timestamps, final disposition, and self-initiation.

The structural witnesses remain separate. Public documentation leaves the derived clocks, ordering, multiplicity, overwrite behavior, and historical continuity unresolved. Calls holding, realized unit availability, timestamped priority histories, applicable callback or fallback states, and public-internal dispatch lineage require internal aggregate evidence.

Table 4. Post-Ida public enrichment and the remaining structural boundary

Object	Public improvement by 2025–2026	Remaining boundary
Internal/public dispatch bridge	more public analytical surfaces and a 2025 CAD–EPR item link	qualified internal dispatch event and field-lineage map needed
Queue and availability	endpoint durations and response-time summaries	calls-holding stock, unit-status history, and realized availability needed
Priority	initial/recorded endpoints, change flags, and clearer current policy	ordered event history, actor, reason, and overwrite trail needed
Continuity and fallback	current policies describe alternate routes and policy-defined callbacks	realized uptime, fallback activation, and applicable callback-execution state needed

The current-data audit illustrates endpoint selection. The 2025 file contains 152,701 records with an arrival field but no dispatch field; the 2026 snapshot contains 97,573. A dispatch-to-arrival duration excludes these arrival-observed records. Dispatch-field completeness is approximately 41.8 percent in both years. Exact initial-versus-recorded priority disagreement is 38.7 percent in 2025 and 37.1 percent in the 2026 snapshot. These all-call values establish pervasive contemporary public-field

missingness and endpoint change. Their scope is the observed audit periods.

The replay sharpens the Ida lesson. Public enrichment improves description; performance measurement still requires lineage, complete clocks, and common denominators. The DV application now carries those requirements through the reporting-to-follow-up chain.

8 Implications for a reported-DV performance study

The framework now turns to the substantive objective. Candidate DV measures begin with the public fields tested under Ida and extend to reports and follow-up when retained links permit. The application derives each measurement status from the evidence required at its stage.

8.1 The shared CAD layer and the reporting-to-follow-up chain

The project’s M7C selection map follows a reported DV-related call through priority, public dispatch and arrival fields, recorded disposition, Electronic Police Report (EPR) production, and possible DVU or AIR entry. The frozen machine-readable records are `M7C_DV_STAGE_MAP.json` and `M7C_FROZEN_SOURCE_INVENTORY.json`:

reported DV-related CAD → priority → public dispatch/arrival fields
→ recorded disposition → EPR → DVU/AIR.

Each arrow requires a retained link, and each downstream stage creates a new denominator. A DV label defines the first box. DVU audit compliance describes sampled DVU files; AIR describes an advocacy-eligible pathway after program launch. Upstream lineage is needed to express either stage as a share of reported DV-related CAD records.

8.2 Reported demand begins at the CAD denominator

The workload estimand begins with calls created in CAD, D_t^{CAD} . It therefore measures reported demand recorded by the dispatch system. Help-seeking attempts that never reach the system, contacts received without a retained CAD record, and underlying DV need occupy earlier stages of the demand chain. Measuring those stages requires external access, survey, hotline, or validation evidence.

This distinction also governs interpretation of call volume. A change in released counts can reflect help-seeking, access, CAD creation, classification, retention, or the underlying phenomenon. R13.0 retains the R12 system-level Ida record-state estimates; it reports no DV call-count trend. The DV application begins at the declared CAD denominator and asks how far performance measurement can proceed from there.

8.3 Classification defines the first denominator

Classification defines the first denominator. The available map lacks a fully qualified, period-specific frozen signal list. A declared rule therefore places records into three groups: DV-related, other, and unresolved. The unresolved group enters the bounds directly.

Let $\mathcal{L}$ denote the declared labeling rule and let $C_i^{\mathcal{L}} \in \{1, 0, U\}$ denote its three states: labeled DV-related, labeled not DV-related, and unresolved. If $N_1 = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = 1\}$ and $N_U = \sum_i \mathbf{1}\{C_i^{\mathcal{L}} = U\}$, the sharp count bound is

$$N_{\text{DV}}^{\mathcal{L}} \in [N_1, N_1 + N_U]. \quad (20)$$

For a fixed CAD denominator N , the corresponding share lies in $[N_1/N, (N_1 + N_U)/N]$. The superscript ties the estimand to rule $\mathcal{L}$. Appendix Section 6.8 gives the labeling semantics and sharpness argument.

At the disposition stage, GOA is a recorded administrative branch. It becomes performance-relevant when policy attaches a specific downstream duty, such as report production or an applicable callback; completion of that duty supplies the corresponding outcome.

8.4 Response-time selection and sharp full-denominator bounds

Let $E_i = 1$ when the timestamps required for a declared response-time clock are observed. Among records labeled DV-related under $\mathcal{L}$, define

$$\pi_E = \Pr(E = 1 \mid C^{\mathcal{L}} = 1), \quad F_E(a) = \Pr(\tau \leq a \mid C^{\mathcal{L}} = 1, E = 1). \quad (21)$$

The public file directly supports the selected distribution F_E . The desired distribution over the full declared-rule denominator is

$$F_{\tau}(a) = \pi_E F_E(a) + (1 - \pi_E) F_0(a), \quad (22)$$

where F_0 is the unobserved distribution for endpoint-missing records. Allowing $F_0(a)$ to range over $[0, 1]$ gives the sharp pointwise bounds

$$\pi_E F_E(a) \leq F_{\tau}(a) \leq \pi_E F_E(a) + (1 - \pi_E). \quad (23)$$

These bounds state exactly what endpoint selection costs. Qualified endpoint recovery or a defensible restriction on the missing group narrows the interval. Unresolved U labels enlarge the outer set through equation (20).

8.5 Priority histories and administrative follow-through

Initial and recorded priorities reveal endpoint disagreement. If $H_i = 1$ denotes at least one priority change, then

$$\Pr(P^0 \neq P^r) \leq \Pr(H = 1) \leq 1. \quad (24)$$

Ordered priority histories replace this bound with measured transitions, their timing, and their direction.

For follow-through, let $L_i = 1$ indicate retained lineage from the CAD denominator to the declared follow-up stage and let $Z_i = 1$ indicate completion of the specified administrative duty. Define

$$\pi_L = \Pr(L = 1 \mid C^{\mathcal{L}} = 1), \quad p_L = \Pr(Z = 1 \mid C^{\mathcal{L}} = 1, L = 1). \quad (25)$$

The full-denominator completion rate p_Z obeys

$$\pi_L p_L \leq p_Z \leq \pi_L p_L + (1 - \pi_L). \quad (26)$$

When lineage coverage is unavailable, the public-data set is $[0, 1]$. Aggregate retained lineage identifies π_L ; linked completion counts identify p_L and close the remaining gap.

8.6 What current evidence can support

Table 5. Measurement frontier along the reported-DV administrative chain

Class	Candidate measure	Supported interpretation
Directly supportable administrative measure	Rule-defined reported-DV counts and $1/0/U$ bounds; recorded disposition composition; public-field completeness	Functions of released fields on an explicit CAD denominator. The declared rule and unresolved class remain part of the estimand.
Bounded or selected measure	Response times among records with the required endpoints; initial and recorded priority endpoints; CAD-EPR key presence; DVU or AIR measures on their own denominators	Equations 23–26 propagate endpoint and lineage coverage; downstream programs retain their selected denominators.
Unavailable from public CAD	Physical service stages; queue delay; realized unit availability or effective capacity; policy-defined callback completion; response quality; victim safety; true DV incidence; a unique mechanism	Internal states, retained links, or separate outcome evidence are required to define these targets.

The table applies [equation \(13\)](#). Classification uncertainty produces count bounds, endpoint missingness produces selected distributions and full-denominator bounds, and downstream programs define new populations. Retained lineage places these components on a common reported-call denominator.

8.7 Target-specific privacy-conscious witness sets

Minimality is target-specific because each performance measure uses a different denominator, clock, or completion rule. Appendix Section 1.3 formalizes the relevant standard. Building on observational equivalence and partial identification, a witness bundle is sufficient for a declared target when it defines the risk set and contracts the target’s identified set to the prespecified precision. It is inclusion-minimal when removing any member restores excess ambiguity or leaves the target undefined. We call this a target-specific witness sufficiency and necessity analysis. Minimum cost or minimum disclosure are separate implementation criteria and can select among several inclusion-minimal bundles.

Qualified period-specific DV labeling and retained event lineage form the common core. The lineage remains internal: its role is to produce reconciled aggregate cells on the same reported-call denominator. The measure-specific additions are shown in Table 6.

Table 6. Target-specific privacy-conscious witness bundles

Target	Required joint witness	Additional information for a broader target
Reported-DV risk set	Qualified period-specific labeling rule, prespecified treatment of U , and internal lineage capable of producing reconciled aggregate cells	Separate validation evidence if the target changes from a declared administrative label to underlying DV status
Receipt-to-assignment delay	CAD receipt and first qualified assignment times for the same calls, with endpoint coverage on the full denominator	Realized unit availability for capacity-adjusted delay or an operational explanation
Receipt-to-qualified-arrival responsiveness	CAD receipt and qualified operational arrival endpoints for the same calls, with endpoint coverage	Calls-holding and availability histories when the target decomposes total time into queue and travel components
Priority revision	Initial priority and the complete ordered priority history, including event times and direction	Actor and reason fields for process or rule-compliance analysis
Administrative follow-through	Duty eligibility, retained CAD-stage lineage, and completion status or time for the applicable downstream stage (EPR, program referral, or policy-defined callback)	Content or quality evidence when the target extends beyond completion of the declared duty

The required evidence is joint rather than marginal. A privacy-conscious release must preserve cells such as reported-DV calls by endpoint coverage and duration bin, or eligible calls by lineage and completion status. Public event identifiers are unnecessary when the agency constructs those cells internally and applies small-cell suppression; the phrase privacy-conscious does not assert a formal privacy guarantee.

The necessity arguments clarify the boundary. Removing the labeling rule changes the reported-DV risk set. Removing lineage permits the same marginal clocks or completion totals to be matched to different CAD calls. Removing one response clock leaves duration undefined; incomplete endpoint coverage leaves a selected distribution and the bounds in [equation \(23\)](#). Endpoint priorities permit offsetting changes, whereas an ordered history measures them. Follow-through requires both duty eligibility and completion on the common denominator.

Calls-holding transitions therefore belong to the minimum bundle for queue delay, while realized unit availability belongs to a capacity-adjusted target. A fully observed receipt-to-qualified-endpoint distribution uses neither. These target-complete bundles support performance measurement conditional on reported DV-related calls received by CAD. Population incidence, victim safety, and mechanism remain distinct estimands.

8.8 A DV-specific estimator would be a new empirical extension

A DV-subset version of the ten-bin event-reference estimator is a separate empirical extension. It requires qualified period-specific classification authority, prespecified treatment of U records, support and cell-size floors, prespecified reference eligibility, and selection handling for endpoints and downstream stages. Its reference set would be requalified for the DV estimand. R13.0 reports

only the frozen R12 system-level estimates.

9 Implications for resource allocation and civilian oversight

The witness analysis turns the measurement frontier into a release design. The same aggregate evidence serves two users. Police managers need common-denominator flows to align capacity with reported demand and locate delay. Civilian oversight needs the same flows to distinguish congestion, prioritization, endpoint coverage, and administrative follow-through.

Table 7. A shared measurement layer for management and oversight

Aggregate object	Resource-management use	Civilian-oversight use
Reported calls by qualified label and priority	Describe incoming workload and priority mix	Reconcile the population used in published measures
Calls holding and realized availability over time	Locate queue pressure and capacity constraints	Separate waiting from endpoint or recording gaps
Receipt, assignment, and qualified endpoint distributions	Compare delay across declared stages and risk sets	Audit endpoint coverage and selected response summaries
CAD-applicable-stage lineage and completion cells	Locate administrative attrition and follow-up workload	Verify completion rates on the original eligible-call denominator

These objects can be released as time-binned joint counts, coverage rates, transition matrices, and duration distributions with small-cell suppression. The design requires neither victim identifiers, unit identifiers, exact locations, nor live deployment information. Stable definitions, versioned labeling rules, and reconciled denominators allow managers and reviewers to reproduce each published measure.

The payoff has two stages. Better releases contract identified sets immediately, making descriptive performance measures more precise and comparable. Richer lineage can then support evaluation of staffing, dispatch, reporting, or policy-defined follow-up reforms when paired with a credible experimental or quasi-experimental design. Ida is unsuitable as an instrument for those effects because it jointly perturbed demand, operations, and recording. Its role is to reveal the measurement problem that a later causal design must first solve.

10 Limitations

The Ida statistic measures rarity within frozen comparison sets; the Kitagawa calculation is an accounting decomposition, and the institutional timeline supplies context. The event jointly stressed operations and record production, so the generating pathway remains set-valued. The 2025–2026 audit describes its observed public-data regime. Help-seeking, access, evacuation, mobility, holidays, weather, COVID-era conditions, redaction, and suppression can also change reported-call populations.

The DV application establishes a measurement frontier. Exact period-specific signal-list authority remains unresolved where noted, and downstream sources condition on selected populations. An empirical DV performance estimate requires qualified classification and the aggregate lineage specified above.

11 Conclusion

Police workload theory links reported demand, effective capacity, priorities, and service. Ida establishes that the public observation of that system can change in an extreme, structured, within-disposition way. Its score is reference-extreme, while the service and record-production paths that generated the pattern remain observationally equivalent.

The measure-specific identified set carries that finding to reported DV-related calls. Declared-rule administrative counts and $1/0/U$ bounds are directly supportable. Endpoint-observed response times describe a selected risk set, with sharp full-denominator bounds determined by coverage. Ordered histories recover priority revision. Retained CAD-to-applicable-downstream-stage lineage places administrative follow-through on the original eligible-call denominator.

Qualified labeling and privacy-conscious aggregate lineage form the common measurement core. Identified-set contraction and witness-necessity arguments then select the clocks, histories, eligibility rules, completion fields, holding stocks, and availability measures required by particular estimands. Released as suppressed aggregate joint cells, these witnesses let managers locate workload pressure and let civilian reviewers audit the same denominators and transitions.

The motivating question therefore has a conditional answer. Public CAD supports defensible administrative measures whose labels, fields, and risk sets are explicit; responsiveness, prioritization, and follow-through require the additional witnesses derived here. Those data would support an empirical DV performance study and stronger future evaluations of resource allocation. True DV incidence, victim safety, and the mechanism behind the Ida pattern remain separate targets.

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